# Technical Design Document (TDD)

# Moffat Bay Lodge Reservation System

## Course: CSD460-300O Capstone in Software Development (2271-DD)

**Project:** Moffat Bay Lodge Reservation System
**Prepared By:**
**Date:**

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### 1. Introduction

#### 1.1 Purpose

The purpose of the Moffat Bay Lodge Reservation System is to provide an online platform that enables customers to view lodge information, search for room availability, make reservations, manage bookings, and communicate with lodge staff. The system will streamline reservation management for administrators while improving customer experience through a user-friendly web interface.

The application will support:

- User account registration and authentication
- Lodge and room viewing
- Reservation booking and cancellation
- Availability checking
- Customer profile management
- Administrative reservation management
- Secure payment processing (future enhancement)
- Customer communication and support

#### Business Goals

- Increase reservation efficiency
- Reduce manual booking processes
- Improve customer satisfaction
- Minimize booking conflicts
- Maintain accurate reservation records
- Provide real-time room availability

**Project Scope**

**Included:**

- Customer registration
- Customer login/logout
- Room browsing
- Reservation management
- Availability search
- Administrative dashboard
- Customer account management

**Not Included (Future Releases):**

- Mobile application
- Loyalty rewards program
- Third-party travel integrations
- Advanced analytics dashboard

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### 1.2 Terminology

| Term | Definition |
|---|---|
| Admin | Authorized lodge staff member who manages reservations and users |
| Availability | Status indicating whether a room can be booked |
| Booking | Reservation made by a customer |
| Check-In Date | Date customer arrives at the lodge |
| Check-Out Date | Date customer departs the lodge |

| | |
|---|---|
| Customer | Registered user making reservations |
| ERD | Entity Relationship Diagram |
| Guest | Visitor using the website without logging in |
| Kanban Board | Agile project management board used to track work |
| Reservation | Confirmed room booking for a date range |
| Room | Lodge accommodation that can be booked |
| Sprint | Time-boxed development period |
| Story Point | Measurement of effort required to complete a user story |
| TDD | Technical Design Document |
| User Authentication | Process of verifying user identity |
| User Story | Requirement written from an end-user perspective |
| QA Testing | Quality Assurance testing process |

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**1.3 User Personas**

**Persona 1: Vacation Traveler**

**Name:** Ruth Johnson

**Age:** 40

**Occupation:** Marketing Manager

**Technical Skill Level:** Intermediate

**Goals:**

- Plan family vacations online
- View room availability quickly
- Complete reservations without calling the lodge
- Receive booking confirmations immediately

**Pain Points:**

- Complicated booking systems
- Unclear pricing information
- Overbooking issues
- Slow website performance

**Needs:**

- Easy navigation
- Secure online reservations
- Instant confirmation emails
- Reservation modification options

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**Persona 2: Returning Customer**

**Name:** Thomas Davis

**Age:** 52

**Occupation:** Business Owner

**Technical Skill Level:** Advanced

**Goals:**

- Quickly book favorite rooms
- Access booking history
- Modify reservations online
- Save personal information

**Pain Points:**

- Re-entering customer details
- Lengthy booking processes
- Lack of reservation history

**Needs:**

- User account access

- Fast reservation management
- Secure profile storage
- Reservation tracking

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**Persona 3: Lodge Administrator**

**Name:** Anna Roberts

**Age:** 46

**Occupation:** Lodge Manager

**Technical Skill Level:** Intermediate

**Goals:**

- Manage reservations efficiently
- Monitor room occupancy
- Update room availability
- Serve customers effectively

**Pain Points:**

- Double bookings
- Difficult record management
- Manual reservation tracking

**Needs:**

- Administrative dashboard
- Reservation reports
- User management tools
- Real-time reservation updates

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**1.4 User Stories**

**High-Priority User Stories**

**US-001**

**Persona:** Vacation Traveler

**Priority:** High

**Story:**

As a vacation traveler, I want to search for available rooms by date so that I can find accommodations that match my vacation schedule.

**Acceptance Criteria:**

- User enters check-in date
- User enters check-out date
- System displays available rooms
- System prevents unavailable room selection

**Estimated Story Points:** 8

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**US-002**

**Persona:** Vacation Traveler

**Priority:** High

**Story:**

As a vacation traveler, I want to complete an online reservation so that I can secure my lodging before arrival.

**Acceptance Criteria:**

- Customer selects room
- Customer enters information
- Reservation is stored

- Confirmation number is generated

**Estimated Story Points:** 13

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**US-003**

**Persona:** Lodge Administrator

**Priority:** High

**Story:**

As a lodge administrator, I want to manage reservations so that I can prevent conflicts and maintain accurate occupancy records.

**Acceptance Criteria:**

- Administrator views reservations
- Administrator edits reservations
- Administrator cancels reservations
- Updates are saved immediately

**Estimated Story Points:** 8

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**Medium Priority User Stories**

**US-004**

**Persona:** Returning Customer

**Priority:** Medium

**Story:**

As a returning customer, I want to create an account so that my personal information can be saved for future reservations.

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**US-005**

**Persona:** Returning Customer

**Priority:** Medium

**Story:**

As a returning customer, I want to log into my account so that I can access reservation history.

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**US-006**

**Persona:** Returning Customer

**Priority:** Medium

**Story:**

As a returning customer, I want to modify an existing reservation so that I can adjust my travel plans when necessary.

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**Low Priority User Stories**

**US-007**

**Persona:** Vacation Traveler

**Priority:** Low

**Story:**

As a vacation traveler, I want to view room details and amenities so that I can choose the best room.

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**US-008**

**Persona:** Lodge Administrator

**Priority:** Low

**Story:**

As a lodge administrator, I want to generate occupancy reports so that I can monitor lodge performance.

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**US-009**

**Persona:** Lodge Administrator

**Priority:** Low

**Story:**

As a lodge administrator, I want to manage customer accounts so that I can assist guests when account issues arise.

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**1.5 Work Estimations (To-Do List)**

**Breakdown of Top 3 User Stories**

**US-001: Search Available Rooms**

| Task | Hours |
|---|---|
| Create room search page UI | 4 |
| Design search form validation | 2 |
| Develop availability search logic | 6 |
| Connect database queries | 3 |

| | |
|---|---|
| Perform unit testing | 3 |
| Perform integration testing | 2 |
| **Total Hours** | **20** |

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### US-002: Online Reservation

| Task | Hours |
|---|---|
| Design reservation form | 4 |
| Create reservation database table | 3 |
| Develop booking functionality | 8 |
| Create confirmation page | 3 |
| Implement reservation validation | 3 |
| Test booking workflow | 5 |
| **Total Hours** | **26** |

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### US-003: Reservation Administration

| Task | Hours |
|---|---|
| Build admin dashboard | 6 |
| Create reservation management UI | 5 |
| Develop edit reservation functionality | 4 |
| Develop cancellation functionality | 3 |
| Security and access control implementation | 3 |
| Test administrator functions | 3 |
| **Total Hours** | **24** |

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**Team Kanban Board Tasks**

The following tasks should be added to the team Kanban board:

**Backend Development**

- Create database schema
- Design user table
- Design room table
- Design reservation table
- Create database relationships
- Implement data access layer
- Develop authentication services
- Develop reservation services

**Frontend Development**

- Create homepage
- Create room search page
- Create room details page
- Create reservation form
- Create Login Page
- Create registration page
- Create customer dashboard
- Create administrator dashboard

**Database Tasks**

- Build ERD
- Create SQL scripts
- Configure database connections
- Populate sample data

**Testing Tasks**

- Write unit tests
- Write integration tests
- Perform system testing
- Perform user acceptance testing
- Conduct accessibility testing
- Execute reservation workflow testing

**Security Tasks**

- Password encryption
- Input validation
- Session management
- Access-control implementation
- SQL injection prevention

- Secure authentication testing

**Documentation Tasks**

- Complete Technical Design Document
- Complete ERD documentation
- Create API documentation
- Create deployment documentation
- Create user guide
- Create test results documentation

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**2. Design**

**2.1 Prototypes**

*To be completed in a later module.*

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**2.2 ERD**

*To be completed in a later module.*

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**3. QA Testing**

**3.1 QA Test Plan**

*To be completed in a later module.*

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**Additional Real-World Requirements Recommended for the Moffat Bay Project (Optional)**

To make this TDD more professional and aligned with industry standards, we can include these considerations during future development:

**Functional Requirements**

- User registration and account recovery
- Email reservation confirmations

- Reservation modification and cancellation
- Room availability updates in real time
- User profile management
- Administrative reporting

**Non-Functional Requirements**

- Page load time under 3 seconds
- 99% application availability
- HTTPS encryption
- Mobile-responsive design
- WCAG accessibility compliance
- Cross-browser compatibility

**Security Requirements**

- Encrypted passwords
- Role-based access control
- Input sanitization
- Secure session handling
- Protection against SQL injection
- Audit logging for administrative actions

**Future Enhancements**

- Online payment processing
- Promotional discount codes
- Customer reviews and ratings
- SMS notifications
- Mobile application
- Business intelligence reporting